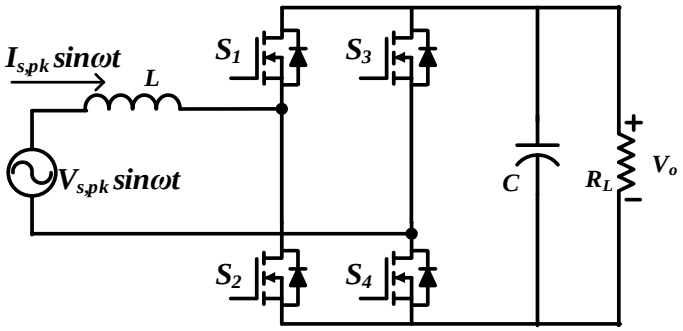
WEEK 4: ASSIGNMENT

1. For the Totempole PFC as shown below:



What will be the RMS current expression for switch S_2? where, symbols have obvious meaning.

A. $\sqrt{\frac{1}{T}\int_0^{\frac{T}{2}} \mathrm{d}(t)(i_s^2(t))\,dt + \int_{\frac{T}{2}}^{T} \big(1-\mathrm{d}(t)\big)(i_s^2(t))\,dt}$

B. $\sqrt{\frac{1}{T}\int_0^{\frac{T}{2}} (1-\mathrm{d}(t))(i_s^2(t))\,dt + \int_{\frac{T}{2}}^{T} \big(d(t)\big)(i_s^2(t))\,dt}$

C. $\frac{I_{s,pk}}{2}$

D. $\frac{I_{s,pk}}{\sqrt{2}}$

Answer: A

2. For the Bridgeless PFC converter fed from 230V, 50 Hz AC mains. What will be the duty ratio of switch S_2 in the negative peak of the input AC voltage, if the output voltage is kept at 400V? (Note: round the result to 3 decimal points). where, symbols have obvious meaning.

Answer: 0.170 to 0.200

3. If a Totempole PFC converter is switching at 70 kHz has perfect unity power factor current drawn is fed from 230V, 50Hz AC input, then output capacitor voltage ripple has following dominant frequency component.

A. 50 Hz
B. 100 Hz
C. 70 kHz
D. 140 kHz

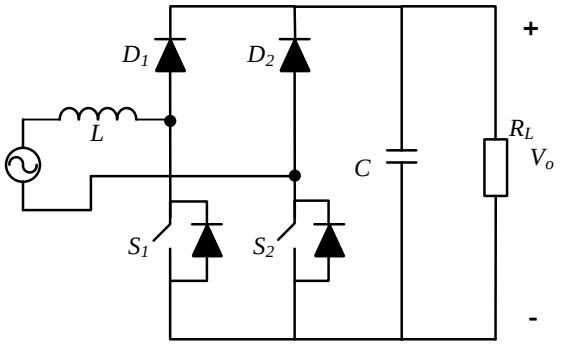
Answer: B

4. For Totempole PFC, which of the following sentence(s) is/are true

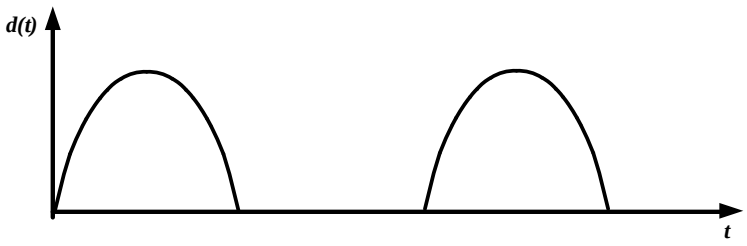
A. The Totempole PFC uses diode bridge rectifier.
B. The Totempole PFC requires minimum of two fast switching devices.
C. The output capacitor sizing is different than that of the single phase boost PFC.
D. The inductor sizing is same as that of single phase boost PFC.

Answer: B, D

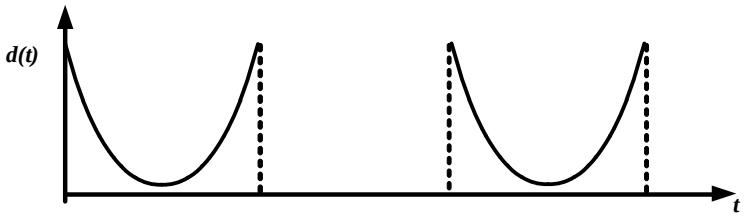
5. For Bridgless PFC, shown below what will be the duty ratio variation of switch S_2 ?



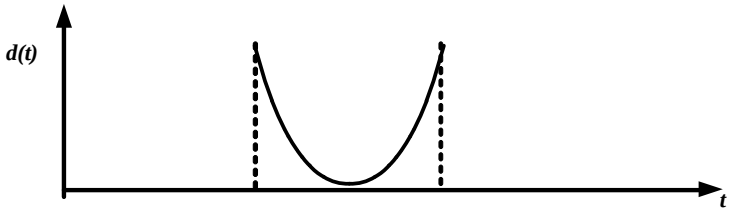
A.



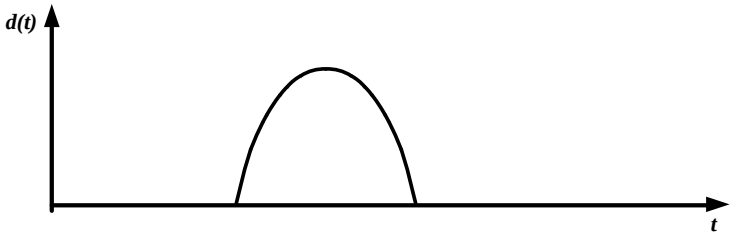
B.



C.



D.



Answer: C

6. At a time, how many devices are in conduction for Totempole PFC converter?
A. 4
B. 3
C. 2
D. 1

Answer: C

7. If the Totempole PFC fed from the 230 V, 50 Hz mains is operating at 80 kHz switching frequency with dual closed loop average current control, then what will be best possible range of current controller bandwidth for achieving the controller objective.
A. $Bandwidth \geq 80$ kHz
B. $Bandwidth < 50$ Hz
C. 40 kHz < $Bandwidth$ < 80 kHz
D. 500 Hz < $Bandwidth$ < 8 kHz

Answer: D

8. For the boost PFC converter what will be the transfer function $G_v(s) = \frac{\widetilde{V_o(s)}}{\widetilde{\iota_l(s)}}$. where, symbols have obvious meaning.

A. $\frac{V_{s,pk}}{V_o}\left[\frac{R_L}{R_L Cs+1}\right]$

B. $\frac{V_{s,pk}}{2V_o}\left[\frac{R_L}{R_L Cs+1}\right]$

C. $\frac{V_{s,pk}}{2V_o}\left[\frac{1}{R_L Cs+1}\right]$

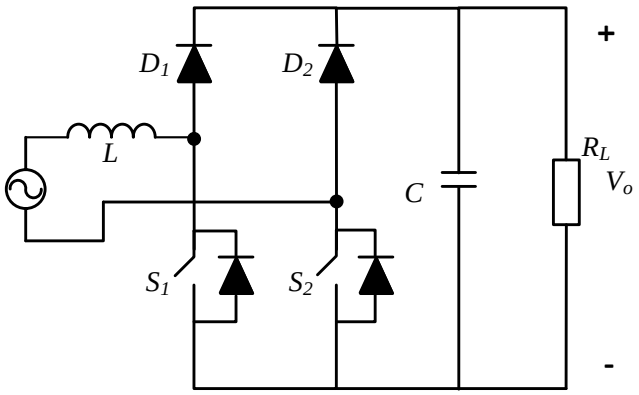
D. $\frac{V_{s,pk}}{V_o}\left[\frac{1}{R_L Cs+1}\right]$

Answer: B

9. For the DCM operation of boost PFC converter, which of the following sentence(s) is/are true:
A. The peak of input current is following the input AC voltage source.
B. While doing the closed loop control, multiplier block and dual closed loop structure is not required.
C. The inductor current becomes zero in every switching cycle.
D. The THD of the drawn current is less.

Answer: A, B, C

10. For Bridgless PFC, shown below which of the following statement is/are true? Where $v_s(t)$ is the input AC voltage.



A. D_1 conducts when $v_s(t) > 0$
B. D_1 conducts when $v_s(t) < 0$
C. D_2 conducts when $v_s(t) > 0$
D. D_2 conducts when $v_s(t) < 0$

Answer: A, D

Answer Key:

1. A	2. 0.170 to 0.200
3. B	4. B,D
5. C	6. C
7. D	8. B
9. A,B,C	10. A,D